

Table 6: The parameters for very large sparse graph. For netgen generator, netgen_8 is a sparse graph with a degree 8, and netgen_lo are variants with lower supplies. Vision we adopt the Vision_inv_05 instance mentioned in Kara & Özturan (2022)

Instance	Instance_Num	Node Num	Arc Num	Total Supply	Capacity_Range	Cost_Range	err	ϵ	d0
Netgen_8	1	1048576	8388608	1024000	[1,1000]	[1,10000]	1e-3	1e-2	1e-4
Netgen_lo_8	1	1048576	8388608	10240	[1,1000]	[1,10000]	1e-3	1e-2	1e-4
Vision	1	3899394	23091149	10000	[0,100]	[9000,11000] <i>capacity</i>	1e-3	1e-2	1e-4

Table 7: The sparsity degree of the solutions

Instance	Cost_Mat	Gurobi_Sol	EOFT_Sol
Node_500_w/o constraint	0.02	0.998	0.996
Node_500_w/ edge+node constraints	0.02	0.999	0.992
Node_5k_w/o constraint	0.002	0.998	0.997
Node_5k_w/ edge+node constraints	0.002	0.982	0.951
Node_10k_w/o constraint	0.002	0.999	0.999
Node_10k_w/ edge+node constraints	0.002	0.962	0.934

D.2 DETAILS ABOUT DATASET

Uniform-MCF: The average out-degree of this dataset is 16, indicating that Uniform-MCF is a high-precision and sparse graph dataset. The Distance matrix here is generated by gaussian function within $[0, 1]^2$ Euclidean space

NETGEN (Klingman et al., 1974) was used to generate random minimum-cost flow, maximum flow, and assignment problems, exported in DIMACS graph format. We use PyNETGEN as a Python implementation of NETGEN in our data generating process.

PyNETGEN are capable of generating minimum-cost network flows problems according to a set of tuneable parameters that control things like the size of the network and the acceptable ranges of arc costs and capacities. It begins by defining source and sink nodes and randomly distributing supply among them. It then generates a set of "skeleton arcs" to create paths from the sources to the sinks. Skeleton arcs are guaranteed to have enough capacity to carry all required flow, ensuring that the problem instance is feasible. After the skeleton is defined, arcs are randomly generated between pairs of randomly-selected nodes until the desired density is reached. The main parameters used for the NETGEN are as follows:

- `nodes` – number of nodes (default 10)
- `sources` – number of source nodes (default 3)
- `sinks` – number of sink nodes (default 3)
- `density` – number of arcs (shown in Table 4)
- `mincost` – minimum arc cost (we set as 10)
- `maxcost` – maximum arc cost (we set as 100)
- `supply` – total supply (we set as 10000)
- `capacitated` – percent of skeleton arcs (0-100) that are capacitated (we set as 100)
- `mincap` – minimum arc capacity (shown in Table 4)
- `maxcap` – maximum arc capacity (shown in Table 4)

E CONVERGENCE OF OFT-SINKHORN ALGORITHM.

Based on the OFT-Sinkhorn algorithm, the corresponding iterative scheme is as follows,